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Fwd: Analytics Global Conference 2025 : Submission (44) has been created.

1 message

Debarshi Mondal <rishimondal99@gmail.com>
To: mayukhs.it.ug@jadavpuruniversity.in

Sun, Jan 26, 2025 at 11:56 PM

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From: **Microsoft CMT** <email@msr-cmt.org>
Date: Tue, Dec 31, 2024 at 8:50 PM
Subject: Analytics Global Conference 2025 : Submission (44) has been created.
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Hello,

The following submission has been created.

Track Name: AGC2025

Paper ID: 44

Paper Title: Deep Learning for Depression Detection Using EEG Signals: A RNN Based Approach

Abstract:

This paper presents a novel methodology for depression detection using Electroencephalogram (EEG) signals from the Multi-modal Open Dataset for Mental Disorder Analysis (MODMA). The proposed approach integrates advanced signal processing techniques with a Recurrent Neural Network (RNN) classification model to objectively identify depressive states. The methodology consists of three key stages: data preprocessing, feature extraction, and RNN based classification. EEG data is preprocessed using bandpass filtering, reshaping, and splitting into training and testing sets, preparing it for deep learning model input. The RNN model is trained on these features, achieving an accuracy of 92.85%, outperforming previous studies that used traditional machine learning techniques like K-Nearest Neighbors (KNN). Results reveal potential challenges, including model overfitting and class imbalance, suggesting avenues for further optimization. This research highlights the capability of deep learning in mental health diagnostics, providing a data-driven approach that may offer earlier, more precise detection of depression. Future work will focus on enhancing diagnostic accuracy by combining EEG with neuroimaging techniques and exploring broader, more diverse patient populations.

Created on: Tue, 31 Dec 2024 15:19:50 GMT

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Primary Subject Area: Analytics methods, tools & techniques

Secondary Subject Areas: Not Entered

Submission Files:

Depression Detection final.pdf (433 Kb, Tue, 31 Dec 2024 15:18:25 GMT)

Submission Questions Response:

1. Paper Type
Full paper (12 pages)
2. Keywords

1/27/25, 12:00 AM

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Depression detection; Electroencephalogram signals; Multi-modal
Open Dataset for Mental Disorder Analysis (MODMA); Recurrent Neural Network (RNN); Deep learning

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